

AZ PROFESSIONAL DEVELOPMENT · FALL 2026

18 talks we'd actually show up for

A speaker brainstorm for the pledge class.
Software, markets, health tech and a couple of wildcards.

```
~/payments - zsh
$ agent "the retry test keeps flaking, fix it"
> reading tests/payments/test_retry.py
> the test sleeps on the real clock and races the timeout
> swapping in a fake clock +12 -4

$ pytest -q
214 passed in 8.31s
✓ opened PR #4817 "Use a fake clock in retry tests"
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```



Service health · us-east-1

Increased error rates · DynamoDB	
Oct 19 · 11:48 PM	Investigating Elevated API error rates for requests to dynamodb.us-east-1
Oct 20 · 12:26 AM	Identified Regional endpoint is resolving to an empty DNS record
Oct 20 · 2:40 AM	Mitigating DNS restored. Dependent services still backed up
Oct 20 · 2:20 PM	Resolved All services recovered
times in PDT · 14 hours, 32 minutes	

Who we're booking for

2/3

want to go into software engineering

1/3

are aiming for finance and quant

A few

are pre-health or into health tech

A talk is worth it if

- 1 There's one real story from inside the company
- 2 The SWE people take notes, not just the finance people
- 3 The speaker sticks around after to talk

Every number in here is current as of September 2026.

Six themes, 18 talks

★ made the top 5



Systems that can't go down

★ The DNS Bug That Took Down Half the Internet

Trillions of Messages, One Migration

Charging a Card Exactly Once



Close to the metal

Rust Won the Kernel
The Kernel Behind Every Chatbot

Your Code Is Slow and the Profiler Knows Why



Building with AI

★ 75% of Google's New Code Is AI-Written

Evals Are the New Unit Tests

Why AI Keeps Getting Cheaper



Markets

★ Microseconds Are Money

★ Betting on Everything

How Market Makers Actually Make Money

Stablecoins Got a Rulebook



Health tech

★ The AI That Writes Your Doctor's Notes

Software the FDA Signs Off On

Your Medical Records Finally Have an API



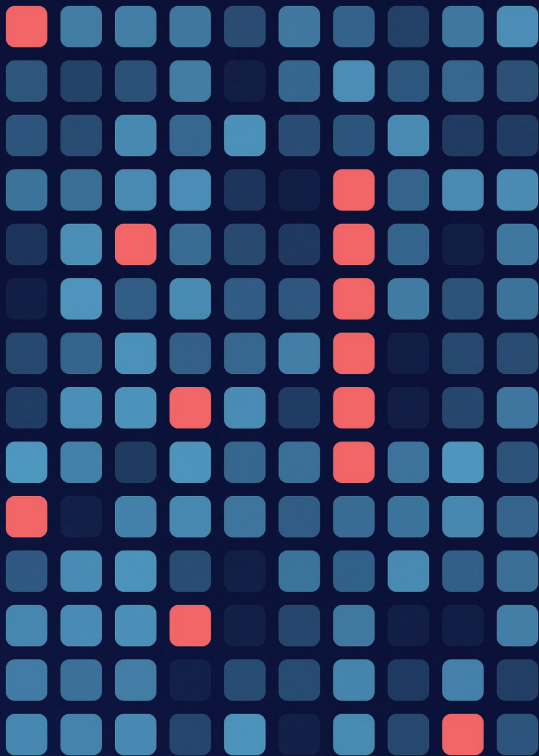
Wildcards

Rebuilding Mario 64 From Scratch

Why Fighting Games Rewind Time

Systems that can't go down

Stories from the teams that get paged at 3am.



★ The DNS Bug That Took Down Half the Internet

One empty DNS record inside AWS took Venmo, Snapchat and Fortnite offline for most of a day.

Who SRE or incident commander at a cloud provider

Ask *What do you check first when everything is red?*

Trillions of Messages, One Migration

Discord moved every message it stores to a new database and went from 177 nodes to 72.

Who Database engineer at a big consumer app

Ask *How do you test a migration you only get to run once?*

Charging a Card Exactly Once

Retries, timeouts and idempotency keys, or why double charges are the scariest bug in fintech.

Who Payments engineer at Stripe, Ramp or Brex

Ask *What's the worst money bug you've seen reach prod?*

Close to the metal

For the people who want to know what the computer is actually doing.



01

Rust Won the Kernel

Linux 7.0 shipped in April with Rust no longer marked experimental. New drivers are being written in it.

Who Systems engineer on Android, Cloudflare or a kernel team

Ask *Where does Rust still get in your way?*

02

The Kernel Behind Every Chatbot

GPU kernels decide how fast and how cheap AI is. Most of the speedup comes from memory, not math.

Who GPU performance engineer at NVIDIA or an AI lab

Ask *How do you know a kernel can't get any faster?*

03

Your Code Is Slow and the Profiler Knows Why

Flame graphs, cache misses and the one-line changes that make things 10x faster.

Who Performance engineer at a trading firm or big tech

Ask *What's the biggest win you got from a tiny change?*

Building with AI

What the job looks like now that models write a lot of the code.

★ 75% of Google's New Code Is AI-Written

That's Sundar Pichai's number from April. So what do engineers actually spend their day on now?

Who Staff engineer who led an AI coding rollout

Ask What do you look for in new grads now?

Evals Are the New Unit Tests

How teams find out a model update quietly broke their product before users do.

Who Applied AI engineer shipping LLM features

Ask What's an eval that caught something humans missed?

Why AI Keeps Getting Cheaper

Kimi K3 has 2.8T parameters but only runs 104B of them per token. Tricks like that keep cutting prices.

Who Inference engineer at Together, Fireworks or Baseten

Ask What's the real bottleneck: chips, memory or power?

The fix is to inject a
fake clock so the test no
longer races the timeout .
214 passed .

Every word here was predicted one token at a time.

Markets

For the finance and quant crowd, and the engineers who build for them.

▼	101.21	692
▲	101.24	364
▲	101.25	185
▼	101.23	455
▼	101.22	706
▲	101.26	637
▼	101.22	656

★ Microseconds Are Money

Light moves faster through air than glass, so firms built microwave towers from Chicago to New Jersey.

Who Low-latency dev or quant dev at a prop shop

Ask *Where's the next microsecond coming from?*

★ Betting on Everything

Kalshi and Polymarket traded \$53B in July, more than 10x what they did last September.

Who Trader on an event-contracts desk

Ask *Where do retail traders give away the most edge?*

How Market Makers Actually Make Money

Spreads, inventory and adverse selection. Being right about the price isn't enough.

Who Options trader at Optiver, IMC or SIG

Ask *What's a trade you were right about and still lost on?*

Stablecoins Got a Rulebook

The GENIUS Act became law in July 2025, and banks spent this year building the rails.

Who Payments or digital-assets lead at a bank or fintech

Ask *Would you run payroll on stablecoins today?*

Health tech

Where software meets patients and the stakes go up.



1

★ The AI That Writes Your Doctor's Notes

Epic launched AI Charting in February. Abridge is already in 300+ health systems.

Who Engineer at an ambient-scribe company

Ask *What happens when it mishears a dosage?*

2

Software the FDA Signs Off On

About 1,500 AI devices are FDA-cleared, most of them reading scans in radiology.

Who ML or regulatory engineer at a medtech company

Ask *How do you update a model the FDA already cleared?*

3

Your Medical Records Finally Have an API

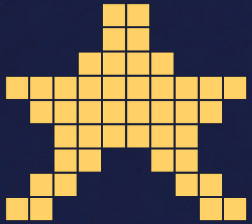
Over 1,000 Epic hospitals are live on TEFCA, and FHIR lets apps pull a patient's full history.

Who Interoperability engineer at Epic or a health-data startup

Ask *Why did this take until 2026?*

Wildcards

Not career advice. Just really fun.

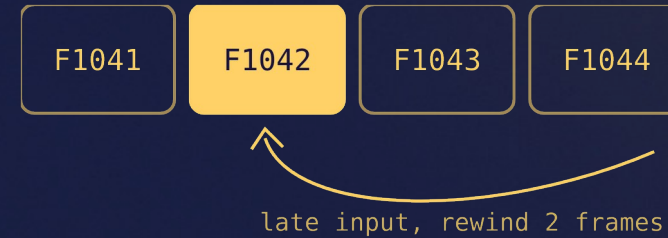


Rebuilding Mario 64 From Scratch

Fans turned the N64 game back into readable C. Now it runs at 60fps on basically anything.

Who Reverse engineer or security researcher

Ask *How do you read code that was never meant to be read?*



Why Fighting Games Rewind Time

Rollback netcode guesses your opponent's input, then quietly rewinds a few frames when it guessed wrong.

Who Netcode engineer at a game studio

Ask *How many frames can you rewind before players notice?*

The top 5 for a mixed room

1

Microseconds Are Money

Markets

SWE and quant people both get something out of it. Hardware, networks and real money on the line.

2

75% of Google's New Code Is AI-Written

Building with AI

Everyone in the room is quietly wondering what this means for their first job.

3

Betting on Everything

Markets

The hottest market of the year, and it's pricing plus exchange engineering in one talk.

4

The DNS Bug That Took Down Half the Internet

Systems that can't go down

Best story on the list. Half the room had an app go down that day.

5

The AI That Writes Your Doctor's Notes

Health tech

Health, AI and a startup vs. incumbent fight, all in one.

Microseconds Are Money

Light moves about 50% faster through air than through glass fiber, so trading firms built chains of microwave towers from Chicago to New Jersey.

```
strategy.log

09:30:00.000000412 md ABC bid 101.25 x 40 ask 101.26 x 3
09:30:00.000000533 md ABC ask 101.26 x 0 level cleared
09:30:00.000000871 sig book imbalance +0.81
09:30:00.000001152 ord BUY 5 ABC @ 101.26 sent
09:30:00.000019304 ack filled 5 @ 101.26
```

tick to trade	wire to exchange
740 ns	18.2 μs

What they'd cover

- Where the time goes: network, kernel, code, exchange
- FPGAs, kernel bypass, and why C++ still runs the floor
- How a firm decides one microsecond is worth millions

Who to book

Low-latency engineer or quant dev at a prop shop like Jump, HRT, Optiver or Citadel Securities

Ask them *Where's the next microsecond coming from?*

75% of Google's New Code Is AI-Written

In April, Sundar Pichai said 75% of Google's new code is now AI-generated. In late 2024 it was about a quarter.

What they'd cover

- What a normal day looks like when an agent writes the first draft
- How code review, testing and on-call have changed
- Which skills got more valuable, and which got cheap

Who to book

Staff engineer who led an AI coding rollout at big tech or a fast-growing startup

Ask them *What do you look for in new grads now?*

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WEATHER · BERKELEY

\$2.1M vol

Will it rain in Berkeley on Big Game day?

Yes

34¢

No

67¢

Order book · Yes

36¢	1,240
35¢	3,905
34¢	8,112
33¢	2,570

TOP 5 · #3 · MARKETS

Betting on Everything

Kalshi and Polymarket traded \$53B in July, up from under \$5B last September. Trading firms now run dedicated desks for it.

What they'd cover

- Pricing a contract that has never traded before
- Making markets on an exchange that's mostly sports
- Matching engines, settlement and the CFTC

Who to book

Trader or quant on an event-contracts desk. Susquehanna was Kalshi's first institutional market maker

Ask them *Where do retail traders give away the most edge?*

The DNS Bug That Took Down Half the Internet

On Oct 20, 2025, a race condition left DynamoDB's us-east-1 endpoint with an empty DNS record. Venmo, Snapchat, Fortnite and Canvas went down with it.

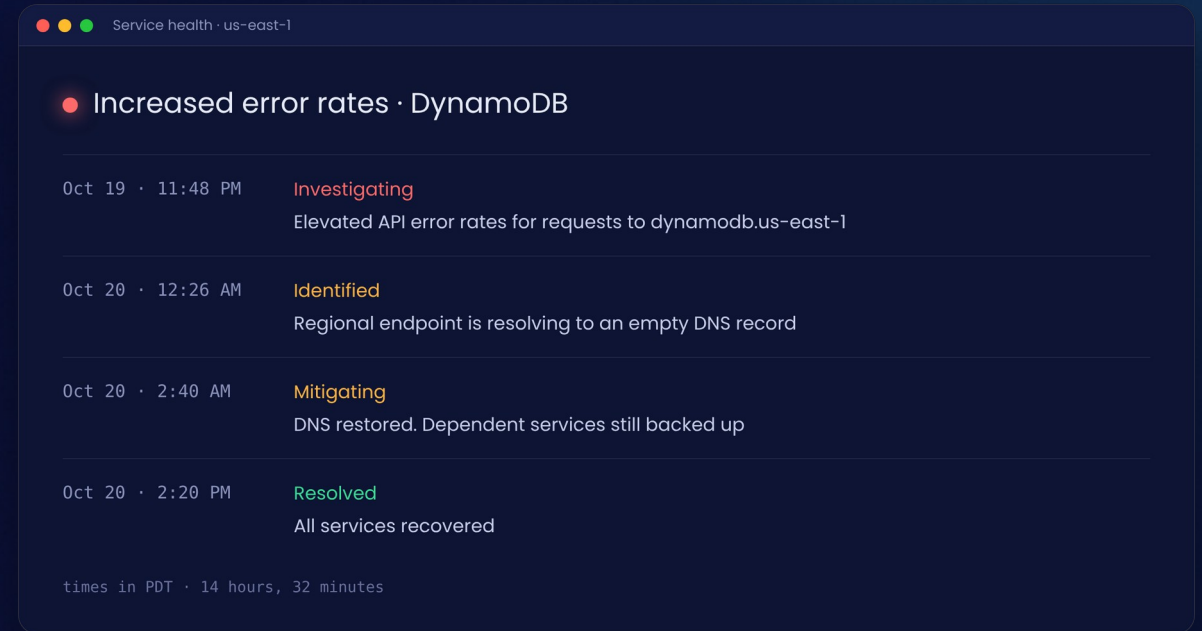
What they'd cover

- How one empty DNS record cascaded for 14 hours
- What an incident call actually sounds like at 3am
- Writing a postmortem that doesn't blame a person

Who to book

SRE or incident commander at a cloud provider or a big consumer app

Ask them *What do you check first when everything is red?*



The AI That Writes Your Doctor's Notes

Epic launched AI Charting in February. Two months later Abridge, already in 300+ health systems, raised at a \$5.3B valuation.

Listening · 06:42

Dr. Patel

Any chest pain when you climb stairs?

Patient

A little tightness, mostly in the mornings.

Dr. Patel

Still taking the lisinopril, 10 milligrams?

Patient

Yeah, every day.

Draft note

AI draft · needs review

Subjective

Intermittent chest tightness on exertion, mostly AM. Adherent to lisinopril 10 mg daily.

Objective

BP 142/88, HR 78. Lungs clear.

Assessment

Exertional chest discomfort, r/o angina. HTN, above goal.

Plan

ECG today. Order stress test. Recheck BP in 2 weeks.

Review & sign

What they'd cover

- Turning a messy 15-minute visit into a clean note
- Keeping a doctor in the loop without slowing them down
- Epic vs. the startups: build, buy or partner

Who to book

ML or product engineer at an ambient-scribe company, or a health system's clinical informatics lead

Ask them *What happens when it mishears a dosage?*

NEXT

How we actually land these people

Start with alumni

LinkedIn → UC Berkeley → Alumni. Filter by company, then by Engineering or Finance.

Ask older actives who they already know. A warm intro gets a yes way faster than a cold DM.

Search LinkedIn like this

```
"low latency" AND FPGA AND Berkeley
```

```
"site reliability" AND Berkeley
```

```
"ambient" AND engineer AND health
```

Find who wrote the blog post

If someone wrote the engineering post, they already have the talk.

Discord, Cloudflare, Stripe, Jane Street, HRT and Abridge all publish engineering blogs with names on every post.

The DM that gets replies

Hi Sam, I'm in AZ at Berkeley. Loved your post on moving your message store to ScyllaDB. Would you do a 30-minute talk for about 40 students who want to work on stuff like this? Zoom or in person, any week this fall works.

Which ones are we booking?

Vote for your top 3 in the form. The most-voted talks get outreach first.

Know someone at one of these companies? Tell us. Warm intros get a yes fastest.

